10.2 Finding Arc Measures

Name _____ Period ____ Date _____

Geometry 10.2 · Student sheet

I can…

- find the measures of minor arcs, major arcs, and semicircles.
- use the Arc Addition Postulate to find arc measures.
- tell whether two arcs are congruent.

Vocabulary · fill in the missing word

Central angle — an angle whose _____ is the center of the circle

Minor arc — an arc whose _____ is less than 180°

Major arc — an arc whose _____ is greater than 180°

Semicircle — an arc whose _____ are the endpoints of a diameter

Congruent arcs — arcs that have the same _____ and are in the same circle or congruent circles

Part 1 · Discovery: angles at the center

Draw a circle and a diameter. Draw one radius above the diameter and one below it. Measure every angle at the center.

Angles above the diameter	Their sum	Angles below the diameter	Their sum	All the angles

A half circle is worth _____°. A whole circle is worth _____°.

Part 2 · Notes and examples (fill in each note, then work the example)

Arc measures: a minor arc has the same measure as its _____ angle. A major arc measures _____ minus its minor arc. A semicircle measures 180°.

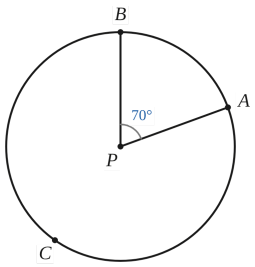
Naming arcs: a minor arc uses its two endpoints; a major arc or a semicircle uses _____ letters.

Example 1

Find the measure of each arc of circle P.

a. arc AB

b. arc ACB



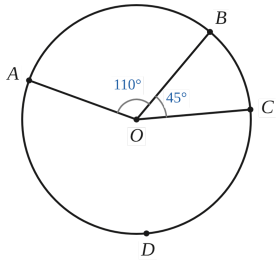
Arc Addition Postulate: the measure of an arc formed by two adjacent arcs is the _____ of the measures of the two arcs.

Example 2

Find the measure of each arc of circle O.

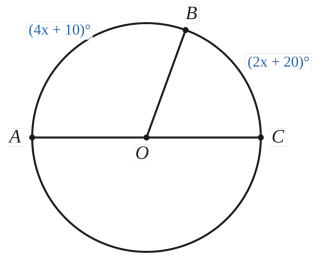
a. arc ABC

b. arc ADC



Example 3

$\overline{AC}$ is a diameter of circle O. Find the value of x. Then find m(arc AB).



Congruent Central Angles Theorem: in the same circle or congruent circles, two minor arcs are congruent if and only if their central angles are _____.

Example 4

Are the arcs congruent? Explain.

a. In circle O, central angles ∠AOB and ∠COD both measure 52°. Is arc AB ≅ arc CD?

b. Circle P has radius 3 and circle Q has radius 5. Each has an 80° arc. Are the arcs congruent?

Example 5

In a survey, 40% of students chose pizza as their favorite lunch. In a circle graph, what is the measure of the arc (and central angle) for pizza?

Part 3 · Partner practice

Partner 1

A central angle measures 125°. Find the measure of its minor arc and of the major arc with the same endpoints.

Partner 2

Three arcs make up a whole circle. They measure (3x)°, (5x + 8)°, and (2x + 12)°. Find x and the measure of each arc.